NEXORA · AI & DATA SCIENCE

Full Proposal

DAMI

دَمي

The right blood unit, to the right hospital, before it's requested... and by drone when there's no time

A Jordanian AI platform that forecasts hospitals' blood needs, rebalances stock before it expires, calls the right donors, and in emergencies launches a cooled medical drone that carries blood from the blood bank to the hospital in minutes.

Team	**NEXORA**
Field	Health · Artificial Intelligence & Data Science · Drones
Competition	Jordan Students Pitch Competition — GEW Jordan
University / Supervisor	.....
Team members	.....

Because blood that arrives late... doesn't arrive.

1 Executive summary

DAMI — A Jordanian AI platform that forecasts hospitals' blood needs, rebalances stock before it expires, calls the right donors, and in emergencies launches a cooled medical drone that carries blood from the blood bank to the hospital in minutes.

Initial investment	Funding needed	Year-3 revenue	Year-3 operating result	Year-3 gross margin
JOD 58,000	**JOD 89,900**	**JOD 325,000**	**JOD 85,250**	**72%**

The prototype is already built as a complete interactive simulator. We start with software in private hospitals to generate early revenue, then run a pilot drone route once permits are granted.

2 The problem

Blood banks face a hard equation: shortages of certain blood types when they are needed, while units expire elsewhere. In major accidents or shortage crises, hospitals need urgent units from the blood bank or another hospital, but vehicles get stuck in traffic or delayed on mountain and remote roads, and in severe bleeding every minute counts.

35–42 days
Shelf life of red blood cells stored at 2–6°C

5–7 days
Shelf life of platelets only, so some are wasted unless distributed smartly

O negative
The type requested first in severe bleeding, before the patient's type is known

Since 2016
Drone blood delivery operates in Rwanda, then Ghana and other countries (Zipline)

- Blood and its components have limited shelf life; some, such as platelets, expire within days.
- Major accidents cause sudden spikes in demand for specific types, especially O negative.
- Drone delivery of blood has been operating worldwide for years; the best-known example is Zipline in Rwanda and Ghana.
- General donation campaigns don't precisely target the types that are short.

3 The solution: three layers in one platform

A three-layer smart platform: (1) it forecasts demand for each blood type at each hospital and rebalances stock between hospitals before it expires; (2) it calls eligible nearby donors by type and last-donation date; (3) in emergencies it launches a medical drone with a smart cooling box (2–6°C) that carries units from the blood bank, or the nearest hospital holding the type, to the hospital in need, with live tracking of temperature, position and ETA.



Forecasting & balancing

A model forecasts 14-day demand for each type at each hospital, warns before shortages, and proposes moving units close to expiry to a hospital that will use them.



Smart donor call-up

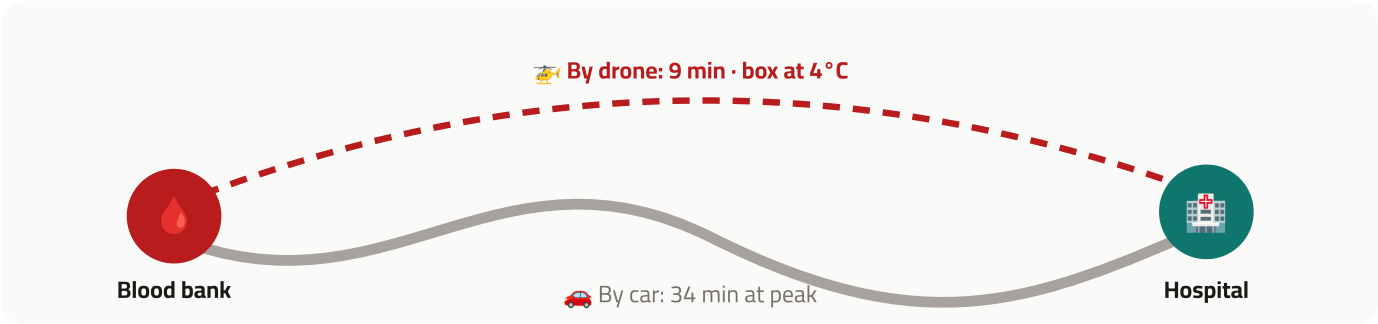
Selects eligible donors (last donation 90+ days ago) who are nearby and most likely to respond, and invites them for exactly the type that's short.



Blood ambulance drone

In emergencies the system picks the nearest source holding the type, compares drone vs car time, and launches a drone with a smart cooling box and live temperature and position tracking.

4 The drone mission: from blood bank to hospital



1. Emergency request
The hospital requests 6 units of O negative in the app with "critical" priority.

2. Source selection
The system checks live stock and picks the blood bank or the nearest hospital holding the type.

3. Launch decision
Comparison: car 34 min at peak vs drone 9 min; the operator approves with one tap.

4. Packing & flight
Units packed in a 2–6 °C cooled box, on a safe route that avoids restricted zones.

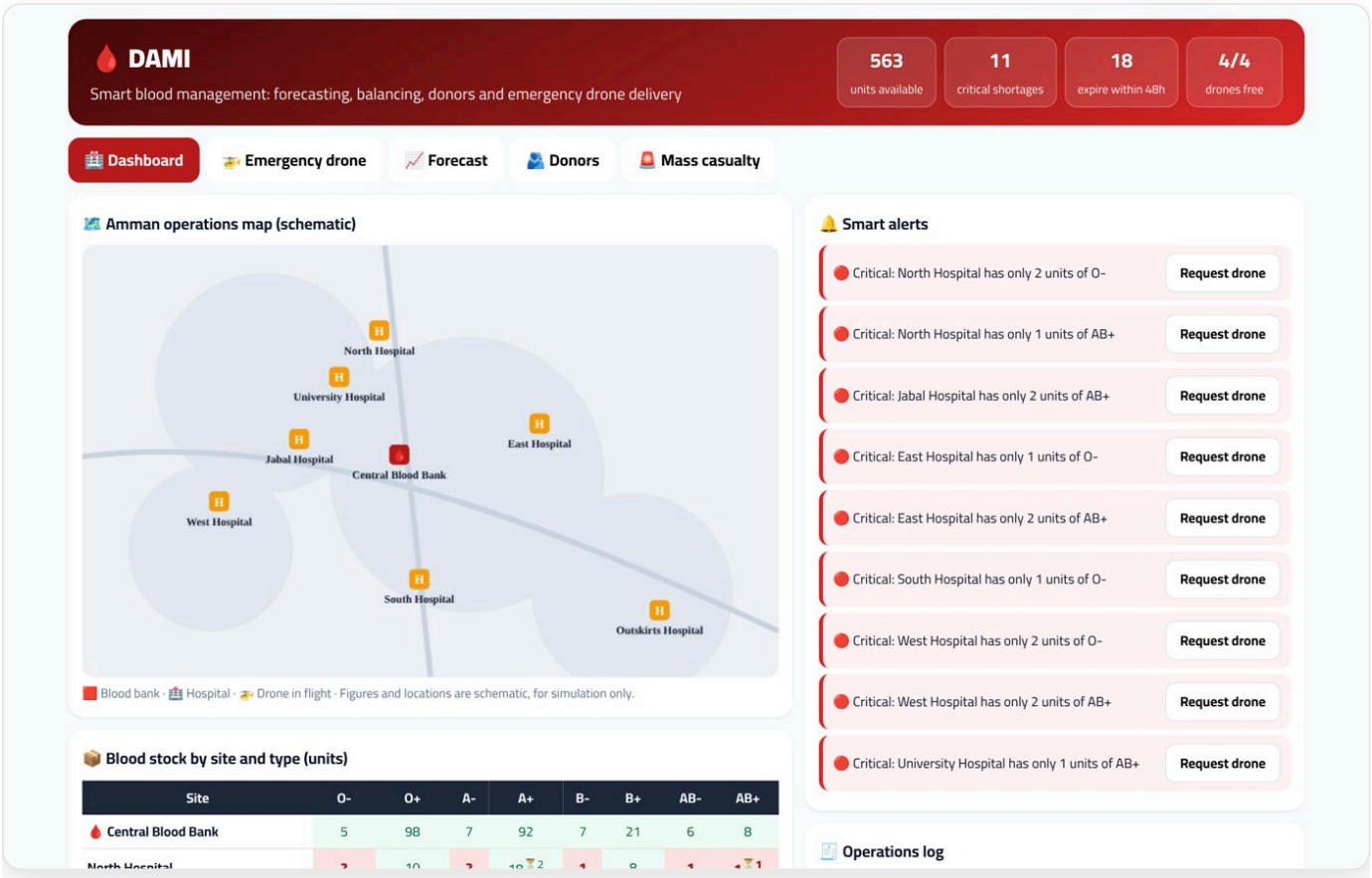
5. Delivery & restock
Landing on the hospital pad, receipt and flight temperature confirmed, and donors called to restock.

Drone and cooling-box specification (pilot)

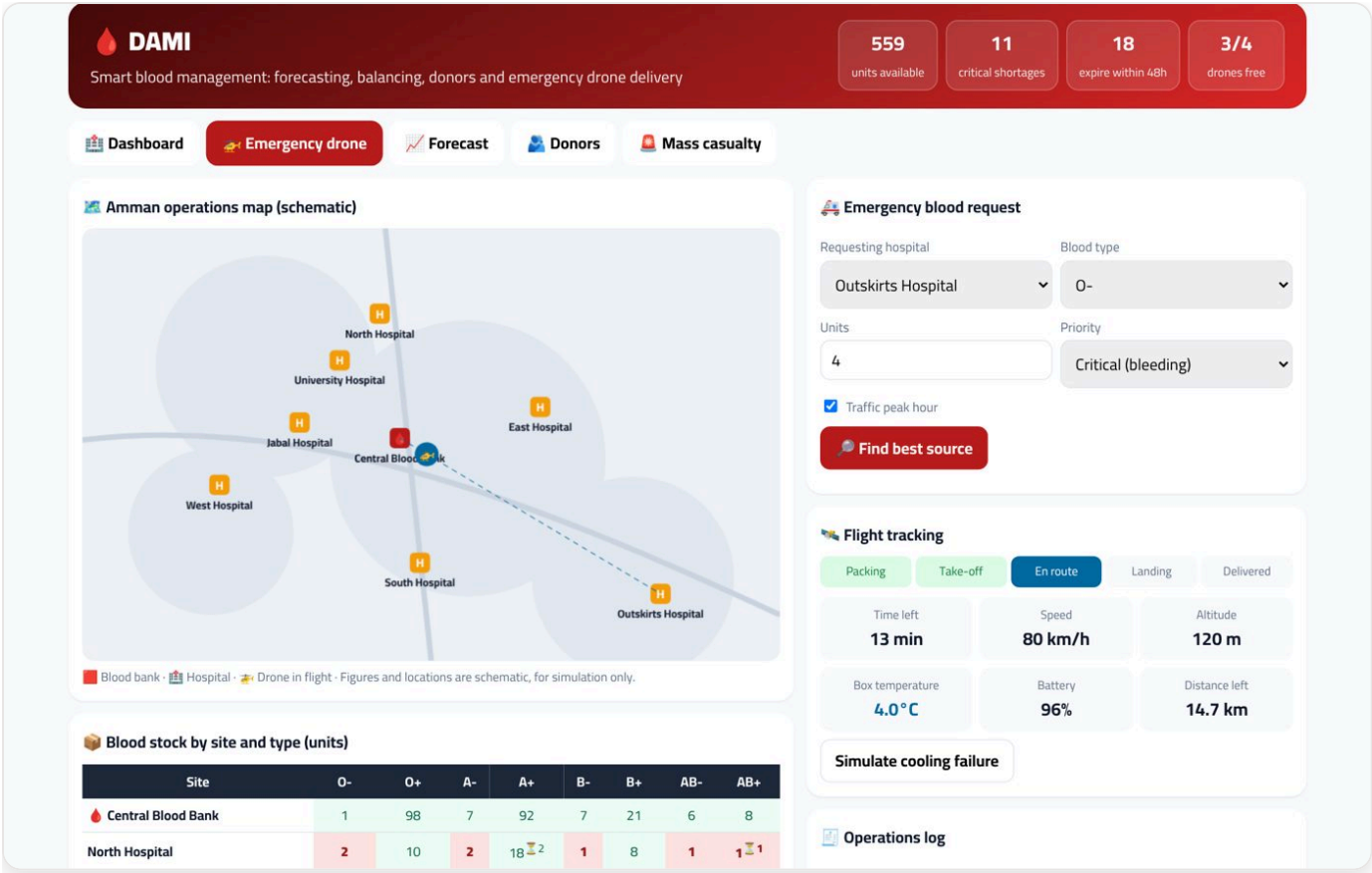
Type	Multirotor drone for the pilot, then fixed-wing VTOL for longer distances
Payload	2–4 kg: 4–6 blood units with the cooled box
Range & speed	15–40 km for urban pilots, at 60–90 km/h
Cooling box	Thermal insulation and phase-change material holding 2–6 °C, with a temperature logger and live alarm
Connectivity	4G/5G cellular with backup link, streaming position, temperature and battery every second
Safety	Emergency parachute, automatic return on link loss, and geofencing
Delivery	Landing on a dedicated pad on the hospital roof or a small-parachute drop in a safe zone
Operations	A licensed pilot monitors from the operations center, on pre-approved routes

⚡ **Live demo for the judges:** A simulated accident on a main road: a hospital requests 6 units of O negative immediately. Within seconds the system picks the blood bank and shows the comparison: car 34 minutes vs drone 9 minutes. The drone takes off on the map, the box holds steady at 4 °C, and eligible donors receive messages to restock.

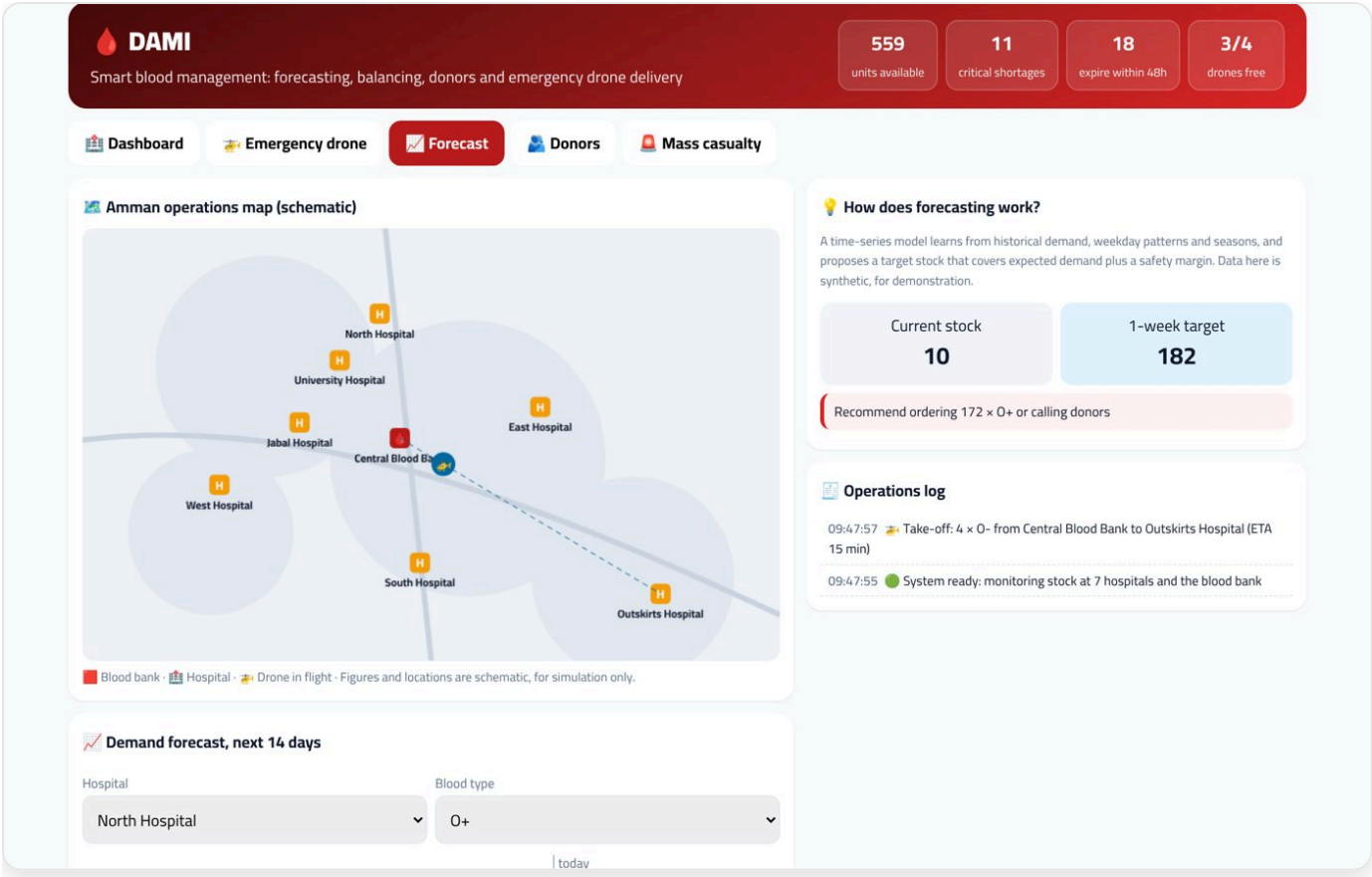
5 The app: prototype screens



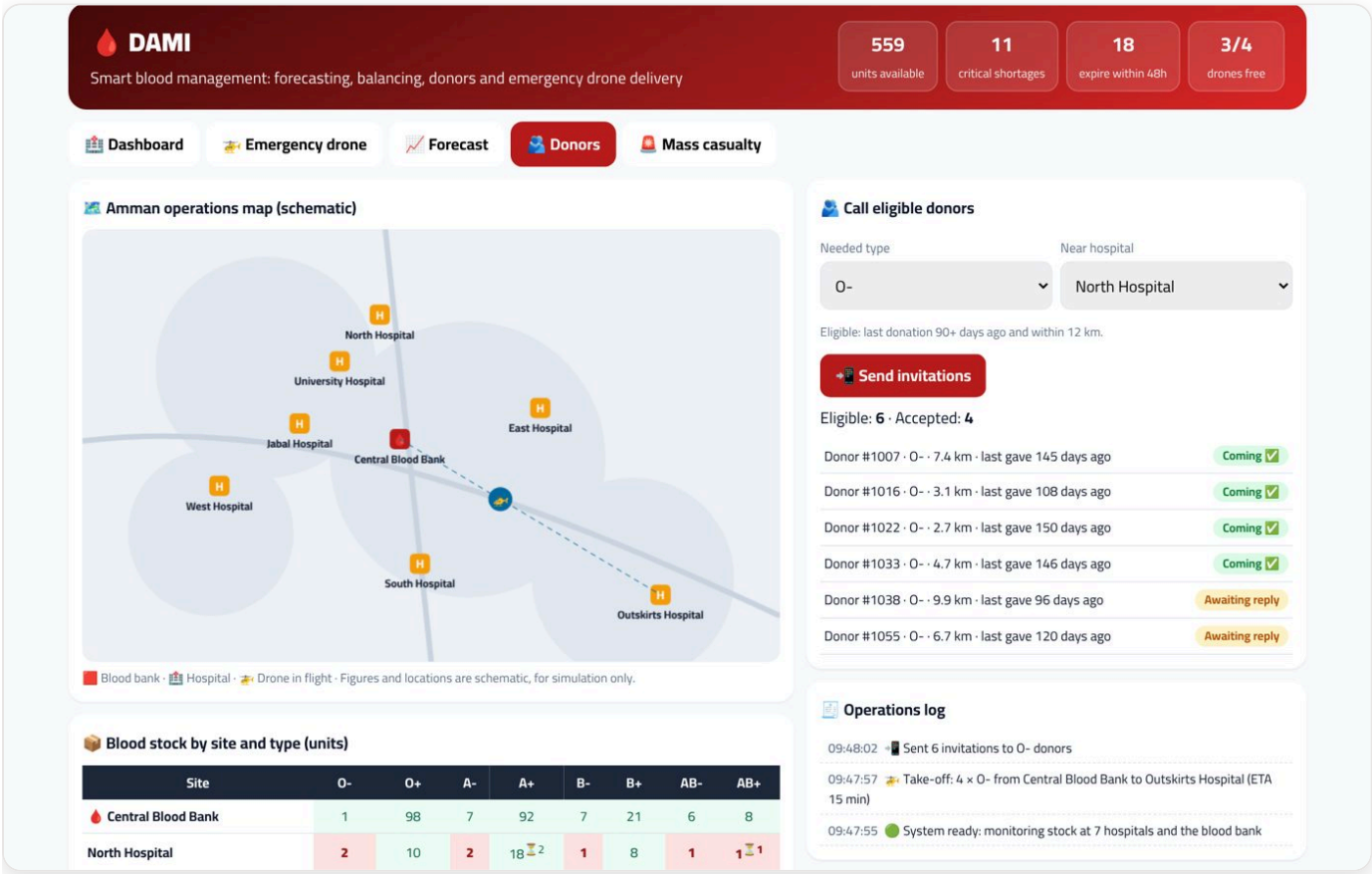
Dashboard: Stock of 8 blood types at the blood bank and 7 hospitals, color-coded shortage and expiry alerts, and one-tap transfer or drone request.



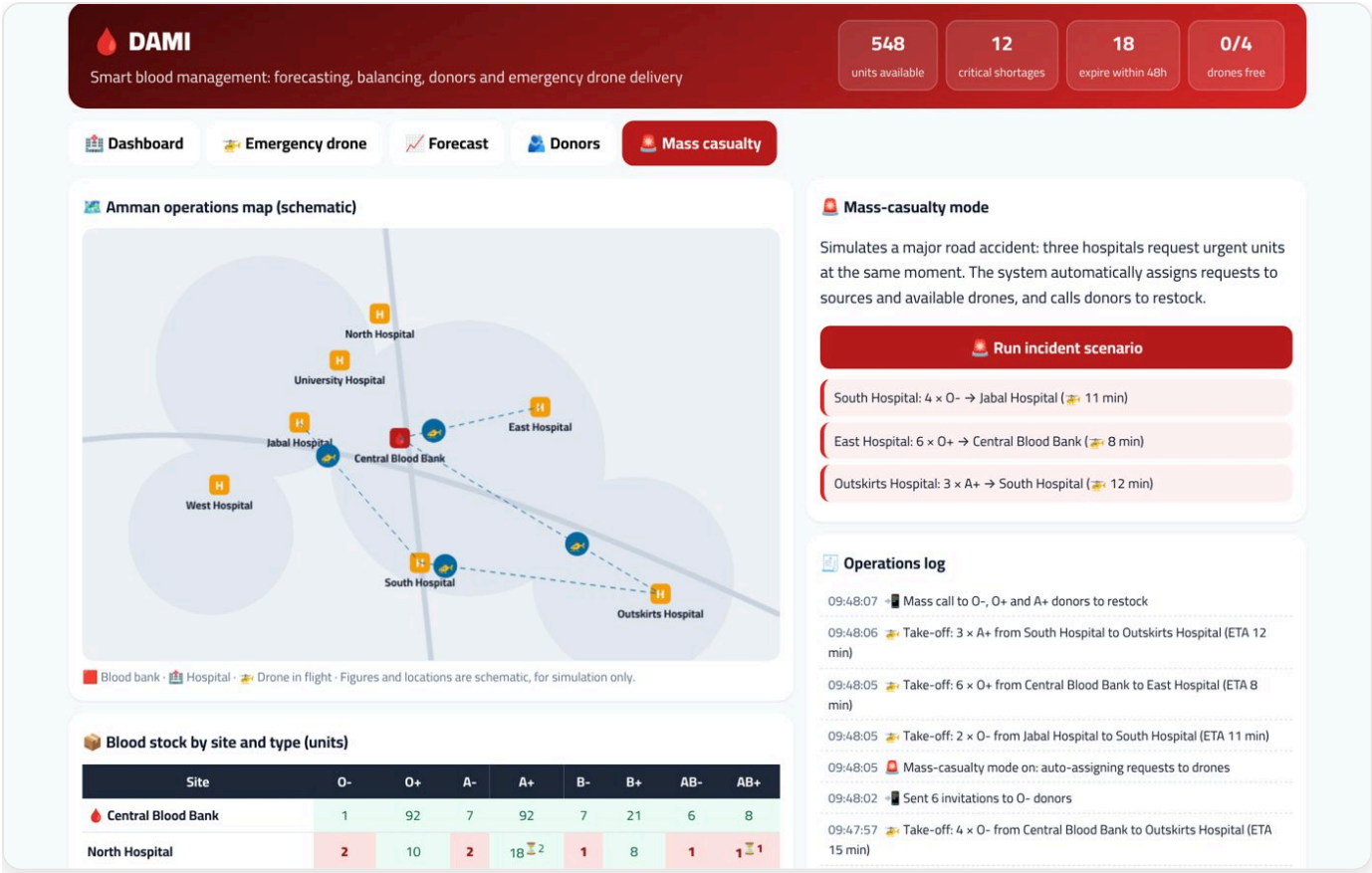
Emergency drone request: Best-source selection, drone vs car time comparison, then launch and tracking: temperature, battery, altitude and distance.



Demand forecast: 14-day demand curve per type and hospital with a confidence band, a target stock and a call-up recommendation.



Donors: Selects eligible donors by type, distance and last donation, sends invitations and tracks replies.



Mass-casualty mode: 3 hospitals request blood at the same moment; the system dispatches drones automatically and calls donors.

6 AI and data

- Time-series forecasting of demand per blood type
- Optimization of stock distribution and best-source selection
- Drone route planning around weather and restricted zones
- Selecting the donors most likely to respond

Data: Blood bank inventories, Request history, Donor database (with consent), Weather and airspace data, Temperature sensor readings from transport boxes

7 Market and competition

Assumption: 3 private hospitals in year 1, then 10, then 20, with drone flights starting in year 2 after licensing.

Alternative	What it offers	Where it falls short
Status quo	Phone calls between blood banks, transport by car, general donation drives	Slow in emergencies, expiry waste, no type targeting
Blood bank management systems	Stock records inside one hospital	No forecasting, no national view, no delivery
Zipline (global)	Drone delivery in Africa and the US	Not present in Jordan per our preliminary research; focused on delivery only
DAMI	**Forecasting + balancing + donors + drone in one platform**	**Built for Jordan; starts with simulation and phases into licensed operations**

8 Business Model Canvas

Key Partnerships

- Ministry of Health & National Blood Bank
- Private and public hospitals, Royal Medical Services
- Medical drone operators and manufacturers
- Civil Aviation Regulatory Commission & security authorities
- Jordan Red Crescent
- Telecom and insurance companies, CSR sponsors

Key Activities

- Building forecasting and balancing models
- Running the emergency ops center and fleet
- Cold-chain and quality management
- Running the donor app and call-up campaigns
- Building partnerships and obtaining permits

Key Resources

- Software platform and AI models
- Drone fleet and smart cooling boxes
- Consented donor database
- Tech team, licensed pilots and a blood bank expert
- Permits and agreements

Value Propositions

- Hospitals: emergency units in minutes instead of half an hour
- Blood banks: less waste by rebalancing before expiry
- Ministry: a live national dashboard and mass-casualty readiness
- Donors: invited at the right time and see their impact
- Patients: blood available when they need it

Customer Relationships

- Annual service contract with an account manager
- 24/7 emergency operations center
- Monthly impact reports
- Donor community, rewards and badges

Channels

- Direct sales to hospital management
- Partnership with the Ministry of Health and the blood bank
- Health conferences and exhibitions
- Media campaigns and universities for donors
- Mobile app and web dashboard

Customer Segments

- Private hospitals (beachhead)
- Ministry of Health hospitals and Royal Medical Services
- National Blood Bank
- Red Crescent and Civil Defense
- Donors (free users)

Cost Structure

- Software and AI development
- Drones and cooling boxes, purchase and maintenance
- Insurance, permits and compliance
- Team and pilot salaries
- Servers, connectivity and marketing

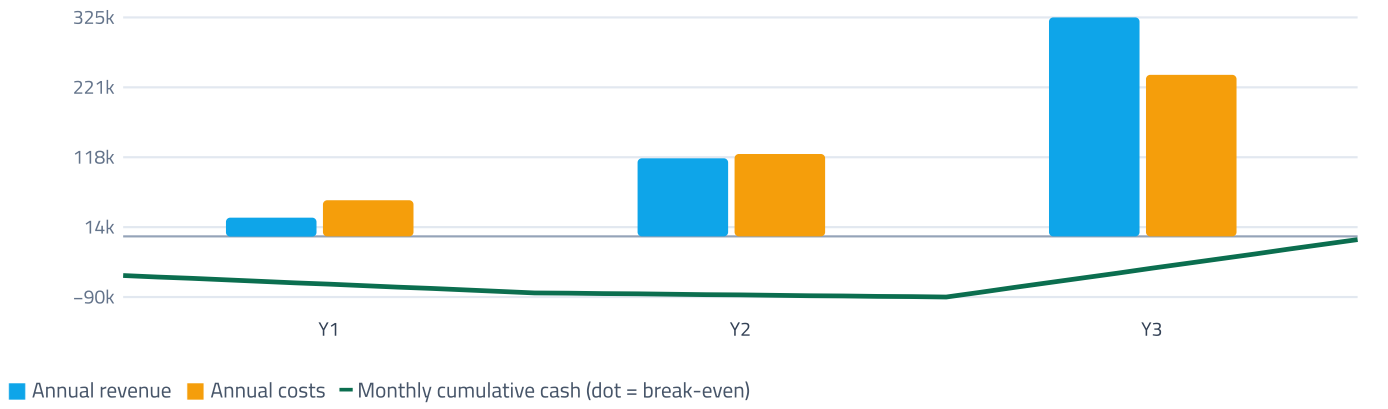
Revenue Streams

- Annual hospital subscription: JOD 6,000
- Emergency drone flight fee: JOD 90
- National dashboard contract: JOD 40,000/year
- Corporate sponsorship of the donor app: JOD 10,000/year

9 Financial feasibility

Initial investment	Funding needed	Year-3 revenue	Year-3 operating result	Year-3 gross margin
JOD 58,000	JOD 89,900	JOD 325,000	JOD 85,250	72%

Item	Year 1	Year 2	Year 3
Revenue	JOD 28,000	JOD 116,000	JOD 325,000
Direct costs	JOD 3,700	JOD 27,200	JOD 89,750
Gross profit	JOD 24,300	JOD 88,800	JOD 235,250
Operating expenses	JOD 50,000	JOD 95,000	JOD 150,000
Operating result	**–JOD 25,700**	**–JOD 6,200**	**JOD 85,250**
Cumulative cash after investment	**–JOD 83,700**	**–JOD 89,900**	**–JOD 4,650**



Scenarios and path to profitability

Scenario	Volume	Costs	Y3 revenue	Y3 result	Funding needed	Break-even
Pessimistic	×0.6	×1.1	JOD 195,000	–JOD 23,850	JOD 173,490	After year 3
Base	×1.0	×1.0	JOD 325,000	JOD 85,250	JOD 89,900	After year 3
Optimistic	×1.4	×0.95	JOD 455,000	JOD 186,850	JOD 71,480	Month 27
Base + grant for pilot fleet	**×1**	**×1**	**JOD 325,000**	**JOD 85,250**	**JOD 59,900**	**Month 33**

Under base assumptions the project does not break even within 3 years because drones and licensing are an early investment; but year 3 is operationally positive (JOD 85,250), and it breaks even in month 27 in the optimistic case. Funding the fleet with a grant lowers the risk.

10 Go-to-market plan

Phase 1 · Months 0–6	Prototype and full simulator (ready) + synthetic data + interviews with 10 blood banks and hospitals.
Phase 2 · Months 6–12	Software pilot in 3 private hospitals at a discounted subscription: forecasting, balancing and donors.
Phase 3 · Months 12–24	Pilot drone route between a blood bank and a hospital after flight permits, with paid flights.
Phase 4 · Months 24–36	National dashboard under a government contract, expanding the fleet to remote governorates and mass-casualty response.

11 Prototype in 12 weeks

Weeks	Deliverables
1–3	Realistic synthetic stock and demand data
4–6	Forecasting, balancing and source-selection models
7–9	Drone simulator and donor app
10–12	Mass-casualty scenario and a drone test inside a closed campus

12 Risks and regulation

Risk	Mitigation
Sensitivity of health data	Formal partnership, encryption and consent
Licensing drone flights in Jordanian airspace	Start with simulation, then a pilot route approved by the Civil Aviation Regulatory Commission and security authorities
Maintaining the cold chain	Medically certified boxes with live temperature monitoring and alarms
Slow government contracting	Start with private hospitals and corporate sponsorship

Work in coordination with the Ministry of Health and the National Blood Bank, complying with blood transfusion, cold-chain and data-privacy standards. Drone operations require permits from the Civil Aviation Regulatory Commission and the relevant security authorities.

13 Impact metrics

- Units saved from expiry
- Minutes saved in emergency delivery
- Share of shipments within safe temperature
- Number of active donors

14 Alignment with national priorities

- Supports health-sector readiness for emergencies and mass-casualty events.
- Aligns with the digital transformation of health services and data protection.
- Reduces waste of a scarce medical resource and improves fair access across governorates.
- Builds Jordanian expertise in medical drones that can be exported regionally.

15 Team

Project Lead / AI Lead	Technical architecture, forecasting models and coordination
Data Scientist	Stock and demand data, and model evaluation
Systems Engineer / Drones & IoT	Drone, cooling box and sensor integration
App Developer	Dashboard, donor app and live demo
Business & Pitch Lead	Health and regulatory partnerships, business model and pitch

16 The Ask

- Seed funding of about JOD 89,900 covering initial investment and the growth period; the pilot fleet (JOD 30,000) can be split out as a health grant.
- A pilot partnership with a blood bank and a private hospital to connect real stock.
- Support on the permit path with the Civil Aviation Regulatory Commission and security authorities.
- Mentoring in the health sector and entrepreneurship.

Financial figures are preliminary planning assumptions set by the team, not market research or supplier quotes, and must be validated before any decision. “Not available in Jordan” means no equivalent was found in preliminary research. The simulator uses synthetic data and schematic locations.